FIFA2026

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Import Data set and remove unnecessary values

Data sourced from Mominul Islam's FIFA-World_Cup-2026-Dataset found here

```
#Imported libraries
library(sqldf)
```

```
Loading required package: gsubfn
```

```
Loading required package: proto
```

```
Loading required package: RSQLite
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
library(stringr)

#Only load necessary data
matches <- read_csv("https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset/raw/refs/hea
```

```
Rows: 104 Columns: 17
-- Column specification --------------------------------------------------------
Delimiter: ","
chr   (2): status, result_type
dbl  (13): match_id, stage_id, venue_id, home_team_id, away_team_id, home_sc...
date  (1): date
time  (1): kickoff_time_utc

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
teams <- read_csv("https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset/raw/refs/heads/
```

```
Rows: 48 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (5): team_name, fifa_code, group_letter, confederation, manager_name
dbl (3): team_id, fifa_ranking_pre_tournament, elo_rating

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
stats <- read_csv("https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset/raw/refs/heads/
```

```
Rows: 1248 Columns: 21
-- Column specification --------------------------------------------------------
Delimiter: ","
chr   (3): player_name, position, data_source
dbl  (14): player_id, team_id, matches_played, matches_started, minutes_play...
lgl   (3): shots, shots_on_target, average_rating
date  (1): last_verified

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
refs <- read_csv("https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset/raw/refs/heads/m
```

```
Rows: 28 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): name, country
dbl (2): referee_id, avg_cards_per_game

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
events <- read_csv("https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset/raw/refs/heads
```

```
Rows: 601 Columns: 6
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): event_type
dbl (5): event_id, match_id, minute, team_id, player_id

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#Remove unwanted columns, this dataset is insanely in depth
str(matches)
```

```
spc_tbl_ [104 x 17] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ match_id              : num [1:104] 1 2 3 4 5 6 7 8 9 10 ...
 $ date                  : Date[1:104], format: "2026-06-11" "2026-06-11" ...
 $ kickoff_time_utc      : 'hms' num [1:104] 19:00:00 23:00:00 19:00:00 23:00:00 ...
  ..- attr(*, "units")= chr "secs"
 $ stage_id              : num [1:104] 1 1 1 1 1 1 1 1 1 1 ...
 $ venue_id              : num [1:104] 1 7 6 3 15 2 10 5 14 11 ...
 $ home_team_id          : num [1:104] 1 3 5 13 7 9 11 15 17 21 ...
 $ away_team_id          : num [1:104] 2 4 6 14 8 10 12 16 18 22 ...
 $ home_score            : num [1:104] 2 2 1 4 1 1 0 2 7 2 ...
 $ away_score            : num [1:104] 0 1 1 1 1 1 1 0 1 2 ...
 $ home_penalty_score    : num [1:104] NA NA NA NA NA NA NA NA NA NA ...
 $ away_penalty_score    : num [1:104] NA NA NA NA NA NA NA NA NA NA ...
 $ status                : chr [1:104] "Completed" "Completed" "Completed" "Completed" ...
 $ result_type           : chr [1:104] "Regular" "Regular" "Regular" "Regular" ...
```

```
 $ home_xg               : num [1:104] 1.84 1.45 1.35 2.76 0.78 1.62 1.25 1.48 4.82 1.95 ...
 $ away_xg               : num [1:104] 0.52 1.12 0.98 0.88 1.54 1.1 0.65 0.72 0.44 1.82 ...
 $ referee_id            : num [1:104] 2 9 11 12 15 1 2 4 5 8 ...
 $ player_of_the_match_id: num [1:104] 16 58 112 332 157 215 293 381 423 524 ...
 - attr(*, "spec")=
  .. cols(
  ..   match_id = col_double(),
  ..   date = col_date(format = ""),
  ..   kickoff_time_utc = col_time(format = ""),
  ..   stage_id = col_double(),
  ..   venue_id = col_double(),
  ..   home_team_id = col_double(),
  ..   away_team_id = col_double(),
  ..   home_score = col_double(),
  ..   away_score = col_double(),
  ..   home_penalty_score = col_double(),
  ..   away_penalty_score = col_double(),
  ..   status = col_character(),
  ..   result_type = col_character(),
  ..   home_xg = col_double(),
  ..   away_xg = col_double(),
  ..   referee_id = col_double(),
  ..   player_of_the_match_id = col_double()
  .. )
 - attr(*, "problems")=<pointer: 0x00000217a00e5590>
```

```
tidy_match <- matches |>
  select(
    match_id,
    home_team_id,
    away_team_id,
    referee_id
  )

str(teams)
```

```
spc_tbl_ [48 x 8] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ team_id                   : num [1:48] 1 2 3 4 5 6 7 8 9 10 ...
 $ team_name                 : chr [1:48] "Mexico" "South Africa" "South Korea" "Czechia" ...
 $ fifa_code                 : chr [1:48] "MEX" "RSA" "KOR" "CZE" ...
 $ group_letter              : chr [1:48] "A" "A" "A" "A" ...
 $ confederation             : chr [1:48] "CONCACAF" "CAF" "AFC" "UEFA" ...
```

```
 $ fifa_ranking_pre_tournament: num [1:48] 14 60 22 40 33 64 56 19 6 7 ...
 $ elo_rating                 : num [1:48] 1810 1620 1800 1740 1795 ...
 $ manager_name               : chr [1:48] "Javier Aguirre" "Hugo Broos" "Hong Myung-
bo" "Miroslav Koubek" ...
 - attr(*, "spec")=
  .. cols(
  ..   team_id = col_double(),
  ..   team_name = col_character(),
  ..   fifa_code = col_character(),
  ..   group_letter = col_character(),
  ..   confederation = col_character(),
  ..   fifa_ranking_pre_tournament = col_double(),
  ..   elo_rating = col_double(),
  ..   manager_name = col_character()
  .. )
 - attr(*, "problems")=<pointer: 0x00000217a00e4e40>
```

```
tidy_team <- teams |>
  select(
    team_id,
    team_name
  )

str(events)
```

```
spc_tbl_ [601 x 6] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ event_id  : num [1:601] 1 2 3 4 5 6 7 8 9 10 ...
 $ match_id  : num [1:601] 1 1 1 1 1 1 1 1 2 2 ...
 $ minute    : num [1:601] 9 9 55 55 67 67 84 86 59 59 ...
 $ event_type: chr [1:601] "Goal" "Assist" "Red Card" "VAR Review" ...
 $ team_id   : num [1:601] 1 1 2 2 1 1 2 1 4 4 ...
 $ player_id : num [1:601] 16 6 39 39 9 25 37 3 85 83 ...
 - attr(*, "spec")=
  .. cols(
  ..   event_id = col_double(),
  ..   match_id = col_double(),
  ..   minute = col_double(),
  ..   event_type = col_character(),
  ..   team_id = col_double(),
  ..   player_id = col_double()
  .. )
 - attr(*, "problems")=<pointer: 0x00000217a00e5060>
```

```r
tidy_events <- events |>
  select(
    event_id,
    match_id,
    event_type,
    player_id,
    team_id)

str(refs)
```

```
spc_tbl_ [28 x 4] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ referee_id        : num [1:28] 1 2 3 4 5 6 7 8 9 10 ...
 $ name              : chr [1:28] "Szymon Marciniak" "Daniele Orsato" "Michael Oliver" "Antho
 $ country           : chr [1:28] "Poland" "Italy" "England" "England" ...
 $ avg_cards_per_game: num [1:28] 4.2 4.8 3.9 4.1 3.6 3.8 4.9 4.5 4 5.1 ...
 - attr(*, "spec")=
  .. cols(
  ..   referee_id = col_double(),
  ..   name = col_character(),
  ..   country = col_character(),
  ..   avg_cards_per_game = col_double()
  .. )
 - attr(*, "problems")=<pointer: 0x00000217a00e5260>
```

```r
tidy_refs <- refs |>
  select(
    referee_id,
    name
  )
```

```r
#Unpivot the matches table for separate home and away team columns
matches_long <- tidy_match |>
  pivot_longer(
    cols = c(home_team_id, away_team_id),
    names_to = "home_away",
    values_to = "team_id"
  )
head(matches_long, 50)
```

```
# A tibble: 50 x 4
   match_id referee_id home_away    team_id
```

```
      <dbl>      <dbl> <chr>          <dbl>
 1        1          2 home_team_id       1
 2        1          2 away_team_id       2
 3        2          9 home_team_id       3
 4        2          9 away_team_id       4
 5        3         11 home_team_id       5
 6        3         11 away_team_id       6
 7        4         12 home_team_id      13
 8        4         12 away_team_id      14
 9        5         15 home_team_id       7
10        5         15 away_team_id       8
# i 40 more rows
```

```
head(tidy_events, 200)
```

```
# A tibble: 200 x 5
   event_id match_id event_type player_id team_id
      <dbl>    <dbl> <chr>          <dbl>   <dbl>
 1        1        1 Goal              16       1
 2        2        1 Assist             6       1
 3        3        1 Red Card          39       2
 4        4        1 VAR Review        39       2
 5        5        1 Goal               9       1
 6        6        1 Assist            25       1
 7        7        1 Red Card          37       2
 8        8        1 Red Card           3       1
 9        9        2 Goal              85       4
10       10        2 Assist            83       4
# i 190 more rows
```

```
#Filter events for only red and yellow cards
tidy_events <- filter(tidy_events, event_type == "Red Card" | event_type == "Yellow Card")
head(tidy_events |> arrange(match_id), 200)
```

```
# A tibble: 47 x 5
   event_id match_id event_type  player_id team_id
      <dbl>    <dbl> <chr>           <dbl>   <dbl>
 1        3        1 Red Card           39       2
 2        7        1 Red Card           37       2
 3        8        1 Red Card            3       1
 4      136       25 Yellow Card         56       3
```

```
 5      137       25 Yellow Card         4       1
 6      139       25 Yellow Card         6       1
 7      142       26 Yellow Card        30       2
 8      143       26 Yellow Card        31       2
 9      144       26 Yellow Card        85       4
10      149       27 Yellow Card       168       7
# i 37 more rows
```

Combine into one df

```
#Select just the match id, home team, away team, and events
combined <- sqldf("
  select m.match_id, e.team_id, e.event_type
  from tidy_match m
  inner join tidy_team t
  on m.home_team_id = t.team_id
  inner join tidy_events e
  on t.team_id = e.team_id
")
```

```
#Check the work
str(combined)
```

```
'data.frame':   103 obs. of  3 variables:
 $ match_id  : num  1 1 1 2 3 4 4 4 4 5 ...
 $ team_id   : num  1 1 1 3 5 13 13 13 13 7 ...
 $ event_type: chr  "Red Card" "Yellow Card" "Yellow Card" "Yellow Card" ...
```

```
head(combined, 100)
```

```
    match_id team_id  event_type
1          1       1    Red Card
2          1       1 Yellow Card
3          1       1 Yellow Card
4          2       3 Yellow Card
5          3       5 Yellow Card
6          4      13    Red Card
7          4      13 Yellow Card
8          4      13 Yellow Card
```

9	4	13	Yellow Card
10	5	7	Red Card
11	5	7	Red Card
12	5	7	Yellow Card
13	6	9	Yellow Card
14	7	11	Yellow Card
15	7	11	Yellow Card
16	7	11	Yellow Card
17	8	15	Yellow Card
18	8	15	Yellow Card
19	8	15	Yellow Card
20	8	15	Yellow Card
21	8	15	Yellow Card
22	14	25	Red Card
23	14	25	Yellow Card
24	15	31	Yellow Card
25	15	31	Yellow Card
26	15	31	Yellow Card
27	17	33	Yellow Card
28	18	35	Red Card
29	18	35	Yellow Card
30	18	35	Yellow Card
31	22	45	Red Card
32	25	1	Red Card
33	25	1	Yellow Card
34	25	1	Yellow Card
35	26	4	Yellow Card
36	27	5	Yellow Card
37	28	8	Yellow Card
38	29	9	Yellow Card
39	31	13	Red Card
40	31	13	Yellow Card
41	31	13	Yellow Card
42	31	13	Yellow Card
43	34	20	Red Card
44	37	25	Red Card
45	37	25	Yellow Card
46	41	33	Yellow Card
47	42	36	Yellow Card
48	47	45	Red Card
49	49	4	Yellow Card
50	50	2	Red Card
51	50	2	Red Card

52	50	2	Yellow Card
53	50	2	Yellow Card
54	51	8	Yellow Card
55	52	6	Red Card
56	52	6	Yellow Card
57	56	14	Red Card
58	56	14	Yellow Card
59	57	20	Red Card
60	58	18	Yellow Card
61	64	30	Yellow Card
62	65	36	Yellow Card
63	66	34	Yellow Card
64	66	34	Yellow Card
65	73	2	Red Card
66	73	2	Red Card
67	73	2	Yellow Card
68	73	2	Yellow Card
69	74	9	Yellow Card
70	78	33	Yellow Card
71	79	1	Red Card
72	79	1	Yellow Card
73	79	1	Yellow Card
74	80	45	Red Card
75	81	25	Red Card
76	81	25	Yellow Card
77	82	13	Red Card
78	82	13	Yellow Card
79	82	13	Yellow Card
80	82	13	Yellow Card
81	85	8	Yellow Card
82	86	15	Yellow Card
83	86	15	Yellow Card
84	86	15	Yellow Card
85	86	15	Yellow Card
86	86	15	Yellow Card
87	89	14	Red Card
88	89	14	Yellow Card
89	90	5	Yellow Card
90	91	9	Yellow Card
91	92	1	Red Card
92	92	1	Yellow Card
93	92	1	Yellow Card
94	94	13	Red Card

```
95         94      13 Yellow Card
96         94      13 Yellow Card
97         94      13 Yellow Card
98         96       8 Yellow Card
99         97      33 Yellow Card
100        99      36 Yellow Card
```

```
#Check the sum of all yellow and red cards
table(combined$event_type)
```

```
   Red Card Yellow Card
         27          76
```

Looking at a paltry 76 yellow cards and 27 red cards for the whole tournament tells us the data set is missing some important info and is incomplete.

So, I found the website FBREF and got the data there instead. They have a very robust system for selecting data so I was able to select only what was needed. Data can be found at this link

```
data <- "
Squad,90s,CrdY,CrdR,2CrdY
dz vs Algeria,4.0,2,0,0
ar vs Argentina,9.0,14,1,1
au vs Australia,4.3,7,0,0
at vs Austria,4.0,2,0,0
be vs Belgium,6.3,10,0,0
ba vs Bosnia-Herz,4.0,4,1,0
br vs Brazil,5.0,7,0,0
cv vs Cabo Verde,4.3,7,0,0
ca vs Canada,5.0,9,2,0
co vs Colombia,5.3,8,0,0
cd vs Congo DR,4.0,8,0,0
ci vs Côte d'Ivoire,4.0,4,0,0
hr vs Croatia,4.0,3,0,0
cw vs Curaçao,3.0,2,0,0
cz vs Czechia,3.0,4,0,0
ec vs Ecuador,4.0,9,0,0
eg vs Egypt,5.3,8,0,0
```

```
eng vs England,8.3,10,0,0
fr vs France,8.0,4,0,0
de vs Germany,4.3,6,0,0
gh vs Ghana,4.0,6,0,0
ht vs Haiti,3.0,4,0,0
ir vs IR Iran,3.0,4,1,0
iq vs Iraq,3.0,2,0,0
jp vs Japan,4.0,7,0,0
jo vs Jordan,3.0,2,0,0
kr vs Korea Republic,3.0,1,0,0
mx vs Mexico,5.0,11,4,0
ma vs Morocco,6.3,10,0,0
nl vs Netherlands,4.3,4,0,0
nz vs New Zealand,3.0,2,0,0
no vs Norway,6.3,3,0,0
pa vs Panama,3.0,3,0,0
py vs Paraguay,5.3,10,0,0
pt vs Portugal,5.0,6,0,0
qa vs Qatar,3.0,3,0,0
sa vs Saudi Arabia,3.0,1,0,0
sct vs Scotland,3.0,4,0,0
sn vs Senegal,4.3,4,1,0
za vs South Africa,4.0,5,1,0
es vs Spain,8.3,20,2,1
se vs Sweden,4.0,2,0,0
ch vs Switzerland,6.7,13,1,0
tn vs Tunisia,3.0,0,0,0
tr vs Türkiye,3.0,2,1,0
uy vs Uruguay,3.0,4,0,0
us vs USA,5.0,11,0,0
uz vs Uzbekistan,3.0,5,0,0"
data_own <- "
Squad,# Pl,90s,CrdY,CrdR,2CrdY
dz Algeria,22,4.0,3,0,0
ar Argentina,24,9.0,15,1,1
au Australia,23,4.3,5,0,0
at Austria,20,4.0,5,0,0
be Belgium,23,6.3,6,1,0
ba Bosnia-Herz,23,4.0,7,1,0
br Brazil,22,5.0,8,0,0
cv Cabo Verde,21,4.3,5,0,0
ca Canada,22,5.0,11,0,0
```

```
co Colombia,21,5.3,8,0,0
cd Congo DR,21,4.0,6,0,0
ci Côte d'Ivoire,22,4.0,4,0,0
hr Croatia,19,4.0,4,0,0
cw Curaçao,22,3.0,7,0,0
cz Czechia,21,3.0,1,0,0
ec Ecuador,21,4.0,8,1,0
eg Egypt,21,5.3,12,0,0
eng England,24,8.3,8,1,0
fr France,22,8.0,6,0,0
de Germany,23,4.3,3,0,0
gh Ghana,22,4.0,6,0,0
ht Haiti,18,3.0,7,0,0
ir IR Iran,20,3.0,6,0,0
iq Iraq,25,3.0,4,1,0
jp Japan,24,4.0,4,0,0
jo Jordan,20,3.0,4,0,0
kr Korea Republic,20,3.0,4,0,0
mx Mexico,25,5.0,4,1,0
ma Morocco,23,6.3,7,0,0
nl Netherlands,22,4.3,3,0,0
nz New Zealand,17,3.0,4,0,0
no Norway,25,6.3,3,0,0
pa Panama,20,3.0,5,0,0
py Paraguay,24,5.3,9,1,0
pt Portugal,22,5.0,7,0,0
qa Qatar,22,3.0,4,2,0
sa Saudi Arabia,18,3.0,6,0,0
sct Scotland,20,3.0,5,0,0
sn Senegal,22,4.3,3,0,0
za South Africa,19,4.0,5,2,0
es Spain,22,8.3,6,0,0
se Sweden,22,4.0,5,0,0
ch Switzerland,23,6.7,8,1,1
tn Tunisia,20,3.0,1,0,0
tr Türkiye,23,3.0,2,0,0
uy Uruguay,17,3.0,5,1,0
us USA,24,5.0,7,1,0
uz Uzbekistan,23,3.0,4,0,0
"

#For the opponent Fairness Score Calculation
```

```
df_op <- read_csv(data)
```

```
Warning: The `file` argument of `read_csv()` should use `I()` for literal data as of
readr 2.2.0.

  # Bad (for example):
  read_csv("x,y\n1,2")

  # Good:
  read_csv(I("x,y\n1,2"))

Rows: 48 Columns: 5
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Squad
dbl (4): 90s, CrdY, CrdR, 2CrdY

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
fair_op_df <- df_op |> mutate(
  fairness = `CrdY`*-1 + `CrdR`*-4 + `2CrdY`*-3)
fair_op_playtime_df <- fair_op_df |> mutate(
  fair_playtime = fairness/`90s`
)
fairness_opponent_FIFA26 <- fair_op_playtime_df |>
  arrange(fair_playtime)

#For own team Fairness Score Calculation
df_own <- read_csv(data_own)
```

```
Rows: 48 Columns: 6
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Squad
dbl (5): # Pl, 90s, CrdY, CrdR, 2CrdY

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
fair_own_df <- df_own |> mutate(
  fairness = `CrdY`*-1 + `CrdR`*-4 + `2CrdY`*-3)
fair_own_playtime_df <- fair_own_df |> mutate(
  fair_playtime = fairness/`90s`
)
fairness_own_FIFA26 <- fair_own_playtime_df |>
  arrange(fair_playtime)

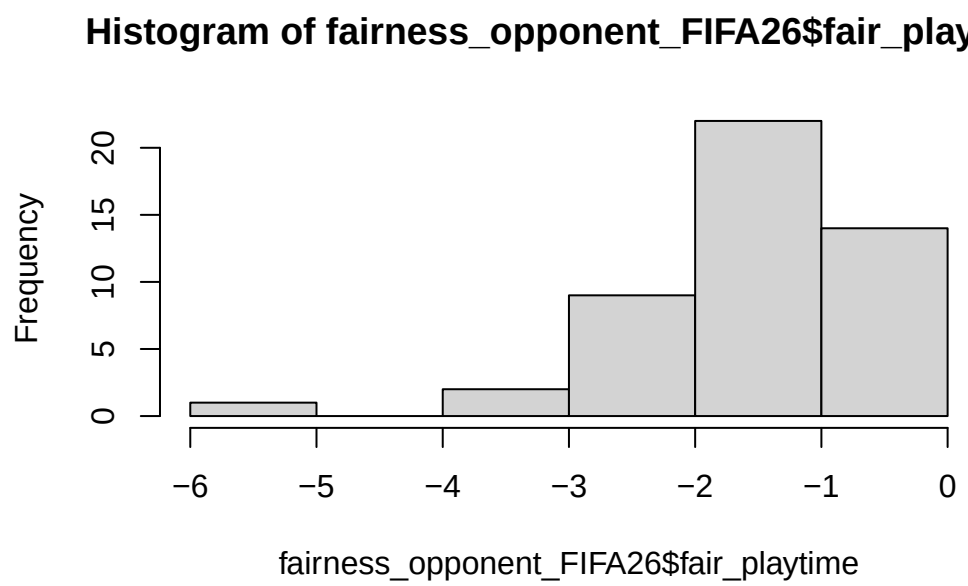
head(fairness_opponent_FIFA26, 50)
```

```
# A tibble: 48 x 7
   Squad              `90s`  CrdY  CrdR `2CrdY` fairness fair_playtime
   <chr>              <dbl> <dbl> <dbl>   <dbl>    <dbl>         <dbl>
 1 mx vs Mexico         5      11     4       0      -27         -5.4
 2 es vs Spain          8.3    20     2       1      -31         -3.73
 3 ca vs Canada         5       9     2       0      -17         -3.4
 4 ir vs IR Iran        3       4     1       0       -8         -2.67
 5 ch vs Switzerland    6.7    13     1       0      -17         -2.54
 6 ar vs Argentina      9      14     1       1      -21         -2.33
 7 ec vs Ecuador        4       9     0       0       -9         -2.25
 8 za vs South Africa   4       5     1       0       -9         -2.25
 9 us vs USA            5      11     0       0      -11         -2.2
10 ba vs Bosnia-Herz    4       4     1       0       -8         -2
# i 38 more rows
```

```r
head(fairness_own_FIFA26, 50)
```

```
# A tibble: 48 x 8
   Squad           `# Pl` `90s`  CrdY  CrdR `2CrdY` fairness fair_playtime
   <chr>            <dbl> <dbl> <dbl> <dbl>   <dbl>    <dbl>         <dbl>
 1 qa Qatar            22   3       4     2       0      -12         -4
 2 za South Africa     19   4       5     2       0      -13         -3.25
 3 ec Ecuador          21   4       8     1       0      -12         -3
 4 uy Uruguay          17   3       5     1       0       -9         -3
 5 ba Bosnia-Herz      23   4       7     1       0      -11         -2.75
 6 iq Iraq             25   3       4     1       0       -8         -2.67
 7 py Paraguay         24   5.3     9     1       0      -13         -2.45
 8 ar Argentina        24   9      15     1       1      -22         -2.44
 9 cw Curaçao          22   3       7     0       0       -7         -2.33
10 ht Haiti            18   3       7     0       0       -7         -2.33
# i 38 more rows
```

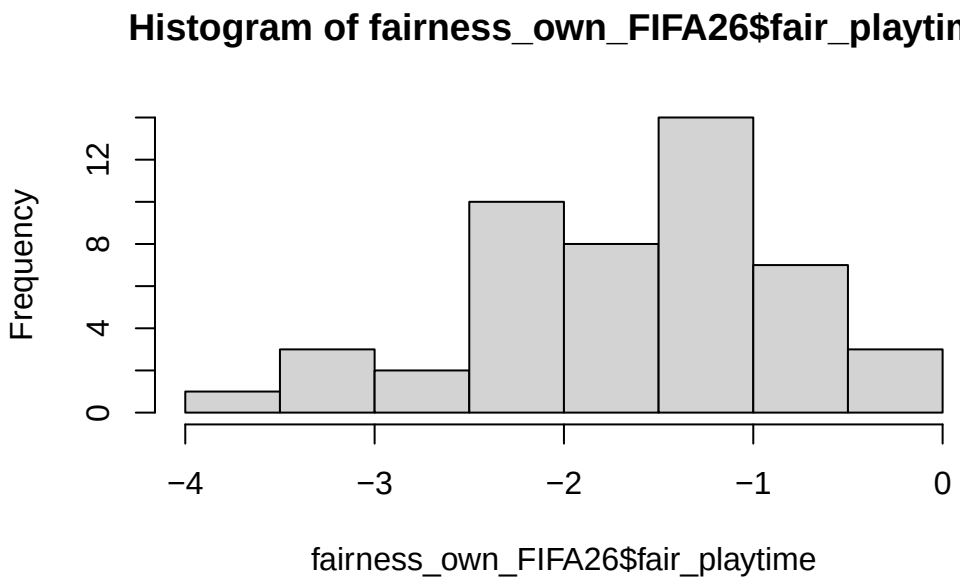
```r
#Plot and run z score test on VS Fairness
hist(fairness_opponent_FIFA26$fair_playtime)
```




```r
analysis_vs <- fairness_opponent_FIFA26 |> mutate(
  fair_mean = mean(fair_playtime),
  fair_sd = sd(fair_playtime),
  fair_z = fair_playtime - fair_mean/fair_sd
) |>
  arrange(fair_z)
head(analysis_vs)
```

```
# A tibble: 6 x 10
  Squad       `90s`  CrdY  CrdR `2CrdY` fairness fair_playtime fair_mean fair_sd
  <chr>       <dbl> <dbl> <dbl>   <dbl>    <dbl>         <dbl>     <dbl>   <dbl>
1 mx vs Mexi~   5      11     4       0      -27          -5.4      -1.50   0.973
2 es vs Spain   8.3    20     2       1      -31          -3.73     -1.50   0.973
3 ca vs Cana~   5       9     2       0      -17          -3.4      -1.50   0.973
4 ir vs IR I~   3       4     1       0       -8          -2.67     -1.50   0.973
5 ch vs Swit~   6.7    13     1       0      -17          -2.54     -1.50   0.973
6 ar vs Arge~   9      14     1       1      -21          -2.33     -1.50   0.973
# i 1 more variable: fair_z <dbl>
```

```r
#Plot and run analysis on Own Fairness
hist(fairness_own_FIFA26$fair_playtime)
```




```r
analysis_own <- fairness_own_FIFA26 |> mutate(
  fair_mean = mean(fair_playtime),
  fair_sd = sd(fair_playtime),
  fair_z = fair_playtime - fair_mean/fair_sd
) |>
  arrange(fair_z)
head(analysis_own)
```

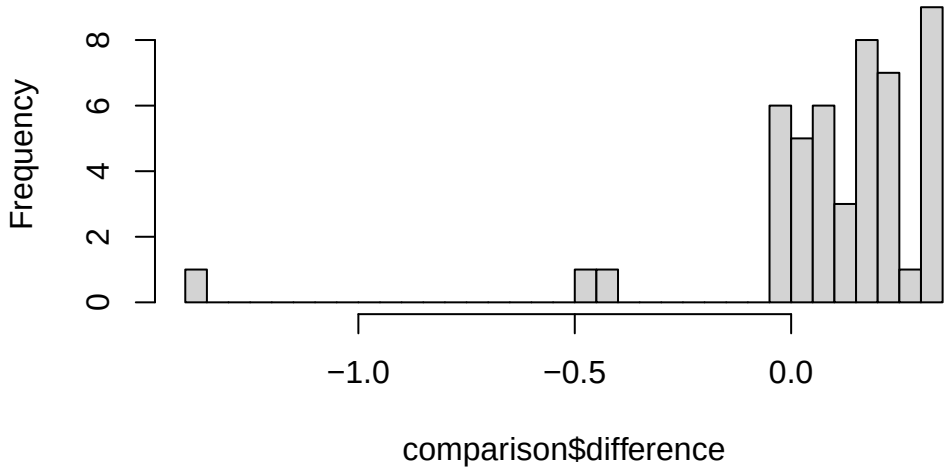
```
# A tibble: 6 x 11
  Squad        `# Pl` `90s`  CrdY  CrdR `2CrdY` fairness fair_playtime fair_mean
  <chr>         <dbl> <dbl> <dbl> <dbl>   <dbl>    <dbl>         <dbl>     <dbl>
1 qa Qatar         22     3     4     2       0      -12         -4        -
1.60
2 za South Af~     19     4     5     2       0      -13         -3.25     -
1.60
3 ec Ecuador       21     4     8     1       0      -12         -3        -
1.60
4 uy Uruguay       17     3     5     1       0       -9         -3        -
1.60
```

```
5 ba Bosnia-H~     23     4     7     1       0      -11         -2.75     -
1.60
6 iq Iraq          25     3     4     1       0       -8         -2.67     -
1.60
# i 2 more variables: fair_sd <dbl>, fair_z <dbl>
```

```
#Calculate difference between own fairness and opponents
comparison <- fairness_own_FIFA26 |>
  mutate(
    difference = fairness_opponent_FIFA26$fair_playtime - fairness_own_FIFA26$fair_playtime
  ) |> select(
    Squad, fair_playtime,  difference
  ) |> arrange(
    difference
  )
head(comparison, 50)
```

```
# A tibble: 48 x 3
   Squad             fair_playtime difference
   <chr>                     <dbl>      <dbl>
 1 qa Qatar                 -4        -1.4
 2 za South Africa          -3.25     -0.485
 3 ec Ecuador               -3        -0.4
 4 co Colombia              -1.51      0
 5 cd Congo DR              -1.5       0
 6 jo Jordan                -1.33      0
 7 kr Korea Republic        -1.33      0
 8 nz New Zealand           -1.33      0
 9 cz Czechia               -0.333     0
10 br Brazil                -1.6       0.0127
# i 38 more rows
```

```
#Plot it and run Z Score Test
hist(comparison$difference, breaks=50)
```




```
analysis_comparison <- comparison |> mutate(
  fair_mean = mean(difference),
  fair_sd = sd(difference),
  fair_z = difference - fair_mean/fair_sd
) |>
  arrange(fair_z, desc=FALSE)
head(analysis_comparison, 50)
```

```
# A tibble: 48 x 6
   Squad             fair_playtime difference fair_mean fair_sd fair_z
   <chr>                     <dbl>      <dbl>     <dbl>   <dbl>  <dbl>
 1 qa Qatar                 -4        -1.4        0.106   0.278 -1.78
 2 za South Africa          -3.25     -0.485      0.106   0.278 -0.867
 3 ec Ecuador               -3        -0.4        0.106   0.278 -0.782
 4 co Colombia              -1.51      0          0.106   0.278 -0.382
 5 cd Congo DR              -1.5       0          0.106   0.278 -0.382
 6 jo Jordan                -1.33      0          0.106   0.278 -0.382
 7 kr Korea Republic        -1.33      0          0.106   0.278 -0.382
 8 nz New Zealand           -1.33      0          0.106   0.278 -0.382
 9 cz Czechia               -0.333     0          0.106   0.278 -0.382
10 br Brazil                -1.6       0.0127     0.106   0.278 -0.369
# i 38 more rows
```

```r
#Remove vs and combine datasets
analysis_vs <- analysis_vs |>
  mutate(Squad = str_remove_all(Squad, "vs "))
head(analysis_vs, 50)
```

```
# A tibble: 48 x 10
   Squad      `90s`  CrdY  CrdR `2CrdY` fairness fair_playtime fair_mean fair_sd
   <chr>      <dbl> <dbl> <dbl>   <dbl>    <dbl>         <dbl>     <dbl>   <dbl>
 1 mx Mexico    5      11     4       0      -27          -5.4      -1.50   0.973
 2 es Spain     8.3    20     2       1      -31          -3.73     -1.50   0.973
 3 ca Canada    5       9     2       0      -17          -3.4      -1.50   0.973
 4 ir IR Iran   3       4     1       0       -8          -2.67     -1.50   0.973
 5 ch Switze~   6.7    13     1       0      -17          -2.54     -1.50   0.973
 6 ar Argent~   9      14     1       1      -21          -2.33     -1.50   0.973
 7 ec Ecuador   4       9     0       0       -9          -2.25     -1.50   0.973
 8 za South ~   4       5     1       0       -9          -2.25     -1.50   0.973
 9 us USA       5      11     0       0      -11          -2.2      -1.50   0.973
10 ba Bosnia~   4       4     1       0       -8          -2        -1.50   0.973
# i 38 more rows
# i 1 more variable: fair_z <dbl>
```

```r
#SQL For Joining all 3 data frames for different metric, own fairness, vs fairness, z scores
Final_FIFA_Fair_MM <- sqldf("
SELECT ao.Squad, c.difference, c.fair_z AS Diff_Fair_Z, ao.fair_playtime AS Own_Fair_Adj, ao
FROM analysis_own ao INNER JOIN analysis_vs av
ON ao.Squad = av.Squad
INNER JOIN analysis_comparison c ON ao.Squad = c.Squad
ORDER BY Diff_Fair_Z")
head(Final_FIFA_Fair_MM, 50)
```

```
              Squad  difference Diff_Fair_Z Own_Fair_Adj  Own_Fair_Z
1           qa Qatar -1.40000000 -1.78202314   -4.0000000 -2.04344736
2    za South Africa -0.48493976 -0.86696290   -3.2500000 -1.29344736
3         ec Ecuador -0.40000000 -0.78202314   -3.0000000 -1.04344736
4        co Colombia  0.00000000 -0.38202314   -1.5094340  0.44711867
5        cd Congo DR  0.00000000 -0.38202314   -1.5000000  0.45655264
6          jo Jordan  0.00000000 -0.38202314   -1.3333333  0.62321930
7  kr Korea Republic  0.00000000 -0.38202314   -1.3333333  0.62321930
8     nz New Zealand  0.00000000 -0.38202314   -1.3333333  0.62321930
9         cz Czechia  0.00000000 -0.38202314   -0.3333333  1.62321930
10         br Brazil  0.01269841 -0.36932473   -1.6000000  0.35655264
```

```
11         mx Mexico  0.01269841 -0.36932473   -1.6000000  0.35655264
12         pa Panama  0.03875969 -0.34326345   -1.6666667  0.28988597
13      sct Scotland  0.03875969 -0.34326345   -1.6666667  0.28988597
14        at Austria  0.05000000 -0.33202314   -1.2500000  0.70655264
15       eng England  0.05043430 -0.33158884   -1.4457831  0.51076950
16       pt Portugal  0.06666667 -0.31535647   -1.4000000  0.55655264
17        be Belgium  0.07786763 -0.30415552   -1.5873016  0.36925105
18        dz Algeria  0.08333333 -0.29868981   -0.7500000  1.20655264
19         fr France  0.08333333 -0.29868981   -0.7500000  1.20655264
20          gh Ghana  0.10000000 -0.28202314   -1.5000000  0.45655264
21     uz Uzbekistan  0.12851406 -0.25350908   -1.3333333  0.62321930
22        cw Curaçao  0.13333333 -0.24868981   -2.3333333 -0.37678070
23         no Norway  0.14285714 -0.23916600   -0.4761905  1.48036216
24      au Australia  0.16279070 -0.21923244   -1.1627907  0.79376194
25     cv Cabo Verde  0.16279070 -0.21923244   -1.1627907  0.79376194
26        ma Morocco  0.18087855 -0.20114459   -1.1111111  0.84544153
27        tr Türkiye  0.19047619 -0.19154695   -0.6666667  1.28988597
28      ar Argentina  0.19444444 -0.18757870   -2.4444444 -0.48789181
29        de Germany  0.19767442 -0.18434872   -0.6976744  1.25887822
30    nl Netherlands  0.19767442 -0.18434872   -0.6976744  1.25887822
31        sn Senegal  0.19767442 -0.18434872   -0.6976744  1.25887822
32       py Paraguay  0.20283019 -0.17919295   -2.4528302 -0.49627755
33    ba Bosnia-Herz  0.21268657 -0.16933657   -2.7500000 -0.79344736
34          es Spain  0.22289157 -0.15913157   -0.7228916  1.23366107
35    ch Switzerland  0.23880597 -0.14321717   -2.2388060 -0.28225333
36        ir IR Iran  0.25000000 -0.13202314   -2.0000000 -0.04344736
37         se Sweden  0.25000000 -0.13202314   -1.2500000  0.70655264
38  ci Côte d'Ivoire  0.25000000 -0.13202314   -1.0000000  0.95655264
39         eg Egypt  0.26415094 -0.11787220   -2.2641509 -0.30759831
40        ca Canada  0.31320755 -0.06881559   -2.2000000 -0.24344736
41          iq Iraq  0.33333333 -0.04868981   -2.6666667 -0.71011403
42   sa Saudi Arabia  0.33333333 -0.04868981   -2.0000000 -0.04344736
43        tn Tunisia  0.33333333 -0.04868981   -0.3333333  1.62321930
44        hr Croatia  0.33333333 -0.04868981   -1.0000000  0.95655264
45          jp Japan  0.33333333 -0.04868981   -1.0000000  0.95655264
46        uy Uruguay  0.33333333 -0.04868981   -3.0000000 -1.04344736
47          ht Haiti  0.33333333 -0.04868981   -2.3333333 -0.37678070
48            us USA  0.33953488 -0.04248826   -2.2000000 -0.24344736
   VS_Fair_Adj   VS_Fair_Z
1   -1.0000000  0.54005759
2   -2.2500000 -0.70994241
3   -2.2500000 -0.70994241
4   -1.5094340  0.03062362
```

```
5   -2.0000000 -0.45994241
6   -0.6666667  0.87339092
7   -0.3333333  1.20672425
8   -0.6666667  0.87339092
9   -1.3333333  0.20672425
10  -1.4000000  0.14005759
11  -5.4000000 -3.85994241
12  -1.0000000  0.54005759
13  -1.3333333  0.20672425
14  -0.5000000  1.04005759
15  -1.2048193  0.33523831
16  -1.2000000  0.34005759
17  -1.5873016 -0.04724400
18  -0.5000000  1.04005759
19  -0.5000000  1.04005759
20  -1.5000000  0.04005759
21  -1.6666667 -0.12660908
22  -0.6666667  0.87339092
23  -0.4761905  1.06386711
24  -1.6279070 -0.08784939
25  -1.6279070 -0.08784939
26  -1.5873016 -0.04724400
27  -2.0000000 -0.45994241
28  -2.3333333 -0.79327575
29  -1.3953488  0.14470875
30  -0.9302326  0.60982503
31  -1.8604651 -0.32040753
32  -1.8867925 -0.34673487
33  -2.0000000 -0.45994241
34  -3.7349398 -2.19488217
35  -2.5373134 -0.99725585
36  -2.6666667 -1.12660908
37  -0.5000000  1.04005759
38  -1.0000000  0.54005759
39  -1.5094340  0.03062362
40  -3.4000000 -1.85994241
41  -0.6666667  0.87339092
42  -0.3333333  1.20672425
43   0.0000000  1.54005759
44  -0.7500000  0.79005759
45  -1.7500000 -0.20994241
46  -1.3333333  0.20672425
47  -1.3333333  0.20672425
```

```
48  -2.2000000 -0.65994241
```

```
#Export the csv
write_csv(Final_FIFA_Fair_MM, "Final_FIFA_Fairness_MM.csv")
```